Evidentiary reasoning with the Analytical Hierarchy Process: Use of graph entropy to rank decision factors

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Abstract

Decision makers often require insight into the relative strength of evidence influencing their choices. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of decision factors' relative evidentiary strength. Pairwise-comparison analysis produces lists of edges for a complete, directed, acyclic graph, often called a *tournament graph*. Pairwise-comparison "preference ratios" provide tournament-graph edge weights, leading to a novel Laplacian matrix. Relative-entropy calculations based on Laplacians for induced subgraphs quantify the relative evidentiary strength associated with the decisional evidence ommitted from them.

Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

> ...I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to *a pretence of exact knowledge that is likely to be false* (emphasis added) (von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision-making represents a formidable research challenge, exposing multiple facets. We conventionally decompose uncertainty into *epistemic* and *aleatory* (or *stochastic*) components (Salicone & Prioli (2018)).

The present study focuses primarily on the epistemic. Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making. Exemplified by Zellweger & Zenger (2023); Rindova & Martins (2024); Ehrig, et al (2025), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

Motivation: How strong is my evidence?

Ehrig, et al (2025) argue that entrepreneurial decision making in particular inescapably requires "...logic and reasoning with the aid of *very small samples of data*" (emphasis added). Circumstances demanding reasoning of this sort arise in many domains beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, entrepreneurial ventures amount to *epistemic* explorations. Howard & Abbas (2016) devote considerable attention to the adverse effects of imperfect information when trying to select optimum scenarios.

Nontrivial challenges inevitably arise *even* when considerable amounts of statistically coherent, phenomenologically illuminating data exist. Cover & Thomas (2006) derive an information-theoretic *data-processing inequality*: "...no processing of Y ... can increase the information that Y contains about X." Corroborating from a statistical-learning perspective, Hastie, Tibshirani, Friedman (2009) define *irreducible variance*. Kay & King (2020) warn of pervasive *nonstationarity* in real-world problems: Statistical-learning training data based on past experience might not represent future conditions very well. More generally, Townsend, et al (2023) articulate the case for "...computationally irreducible barriers to the predictive capabilities and task performance advan-

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tages of AI systems…" for many tasks.

Extending Dempster's (1967) seminal concept, Shafer (1976) offered a profoundly rigorous approach to quantifying epistemic uncertainty. More-recently, Shenoy (2023) applied Shafer's reasoning to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024); Ehrig, et al (2025)).

Hué, et al (2026) use a stochastic-detection area-under-the-curve (AuC) (*cf.* Fawcett (2026)) approach to analyze evidentiary factors' relative strengths. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) offers an emerging approach to uncertainty characterization based on sheaf theory (Rosiak (2022)).

Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for "…logic and reasoning with the aid of very small samples of data" (Ehrig, et al (2025)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability for group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible *evidentiary span*: The range of measurable factors believed to influence the attractiveness of each distinct scenario under consideration. The responsible individual decision maker or panel of experts then performs *pairwise comparisons* to assign a "preference ratio" between each pair of factors within the range (Merrick, van Dorp, Singh (2005)). Illustrated in a case study in a subsequent section, AHP produces the scenario rankings according to attractiveness based upon their respective measurable factors.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the pairwise comparisons as stronger empirical evidence emerges (Li, H., et al (2025)).Li, J. et al (2025) combined back-propagation neural networks with the AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent elicitation of pairwise comparisons from experts. Lukinskiy, Lukinskiy, Bazhina (2025); Sáenz-Royo & Chiclana (2026) studied methods to improve the rubustness of expert-judgement inputs.

Evidentiary-strength analysis using the AHP.

That *a set of AHP pairwise comparisons defines a complete, directed, acyclic graph* (DAG) provides the seminal insight underlying this investigation. Complete DAGs belong to a special class of graphs called *tournament graphs* (Moon (2015)). In general, "the entropy of a graph is interpreted *as its structural information content* and serves as a complexity measure" (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Brown, et al (2020) studied the particular characteristics of tournament graphs.

What questions do entropies of AHP-based tournament graphs allow us to answer? Endogenous to the AHP formulation itself, relative entropy quantifies the marginal informational contribution by a subset of factors within the evidentiary span. This addresses a the objective pursued by Hué, et al (2026). Graph-entropy measures however do not require extensive data comprised of both positive and negative cases as with Hué's approach. Congruent with the AHP methodology explored here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of *belief-function graphical models* to decompose their relative evidentiary strength.

By what mechanism do we infer the differential information between AHP-based tournament graphs with distinct evidentiary spans? Relative entropy provides a well-understood approach (Cover & Thomas (2006)). Relative entropy for graphs has received considerable attention (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Brown, et al (2020) explore entropy differences between same-order tournament graphs with different directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost due to decision factors omitted from the latter.

What innovation is required to define a methodology for AHP-based evidentiary-strength analysis? First, we must formulate an entropy definition for tournament graphs with *weighted* edges. The investigation by Brown, et al (2020) focused on tournament graphs with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022)formulate entropies for regular (non-directional) graphs comprised of weighted edges. The preference ratios from the AHP pairwise comparison provide the weights used here.

Second, relative entropies of hierarchically linked AHP subgraphs must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into contituent factors. The AHP method produces a set of decision-factor pairwise comparisons — and associated tournament graph — for each hierarchic "cluster". (Cover & Thomas , 2006, §2.5) describe *chain rules* for

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entropy, relative entropy, and mutual information. Such mechanisms offer a rigorous approach to link hierarchically adjacent tournament graphs.

Why is "heavy" graph-entropy machinery needed? We might intiutively expect that individual pairwise preference ratios suffice to inform us about the most-influential factors. The AHP's machinery however accounts for "mutual coupling" between decision factors (Saaty (1986)). That pairwise comparisons define a graph — a network — provides the basis of its power. Our graph-entropy machinery accounts for the *aggegate effect* of all the evidence informing a decision or risk assessment. Individual preference ratios fail to provide this.

Course of investigation.

We begin with a brief illustration of AHP methodology. A *real-world* case study concisely demonstrates its application to obtaining a preference-ranked set of scenarios. The actual ranking lies beyond our specific interest. The DAGs resulting from the pairwise preferences provide our focus.

We then calculate the entropy for our pairwise-preference DAG. This extends work by Brown, et al (2020) through construction of a Laplacian that includes the edge weights. Our AHP-graph Laplacian fuses elements of that for a conventional unit-edge-weighted DAG Laplacian with components from the AHP reciprocal matrix for edge weights. This incorporates information about degree of influence in the interconnected evidentiary framework.

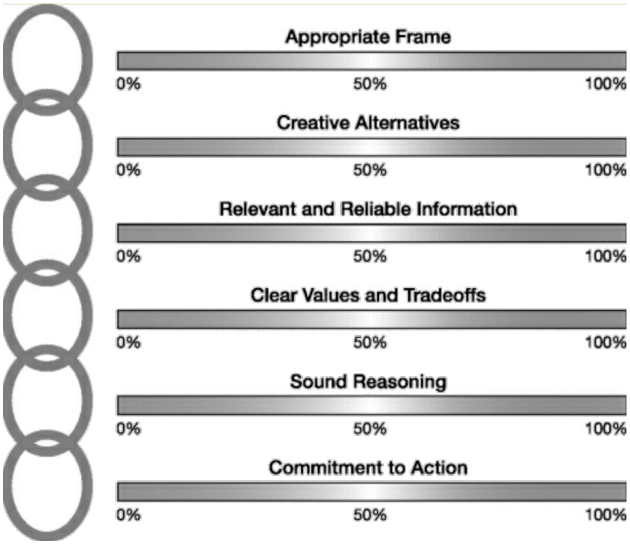
We then calculate entropies and relative entropies from our AHP-based DAG structure and interpret the results for our real-world case study. Approaches demonstrated by Jiroušek, Kratochvíl, Shenoy (2023) guide our thinking. We also explore the relationship between preference ratios relative entroy.

AHP as path to achieving *Decision Quality*.

The pattern of reasoning underlying Saaty's (1986) AHP methodology corresponds to patterns on which prominent applied decision-analysis practices are based. *Decision Quality* (DQ) represents a prominent example (Spetzler, Winter, Meyer (2016)). 1 depicts the DQ framework. It outlines *de facto* standards of practice for the Society of Decision Professionals (SDP), a community of applied decision analysts and scientists. The DQ frameworks contains six *links*:

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with prospcriptive judgements to provide analytical clarity.

Figure 1: Decision-Quality framework. (From Spetzler, Winter, Meyer (2016), Figure 2.1)



- By definition, the act of deciding involves selection of one or more options from a discrete, precisely defined set of *Creative Alternatives*.
- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.
- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified.
- *Sound Reasoning* defines the logic employed in identifying the best alternative. Creighton (1909) cautioned that "...it is not possible to lay down definite rules for the guidance of thinking in every case." The reasoning approach must therefore fit the circumstances resulting from the links above.
- *Commitment to Action* represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to adopt or act upon the best alternative resulting from the links above, a decision arguably has not occurred. The analysis was academic.

Now, the DQ framework defines a scientific, evidentiary-reasoning pattern that is philosophically consistent with the design-of-experiements practice (Cox & Reid (2000)). Its mention here amplifies the theme of scientific reasoning.

Case study in the Analytic Hierarchy Process (AHP).

Bu, et al (2026) employed AHP in making tradeoff de-

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cisions related to the design of a proton exchange membrane fuel cells (PEMFCs), which "...can effentialy convert chemical energy from fuels into electricity without causing polution...". Bu, et al (2026) report device-physics technical details that are tangential to the present exploration. Their use of the AHP methodology primarily interests us.

This case study suits the purposes of this investigation for three reasons.

1. *It demonstrates a real-world application*. Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration.

2. *This case is recent*. Given the AHP literature spans decades, a reader might succomb to the temptation to conclude that the method is anochronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically essential application.

3. *An informative but simple illustration can be extracted*. The AHP problem was designed to be *hierarchical*. Complex MCDM problems often involve dividing and conquering. A large AHP application might involve the "nesting" of smaller AHP formulations, analogous to Russian Matryoshka dolls. Our demonstration considers only one of the "inner" AHP formulations, but illustrates the presence of a broader context.

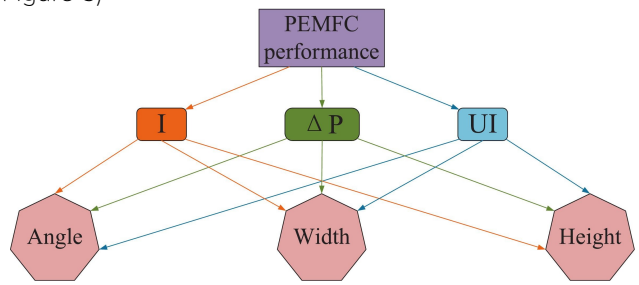
The AHP-methodology description herein employs the terminology of Saaty (1986). Variations in terminology occur througout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a "weight vector" or "weights". This work remains close to Saaty's conventions.

Specification of the evidentiary span.

Figure 2 reproduces the "Hierarchical structure model" employed by Bu, et al (2026) for their fuel-cell design. Explicitly representing the decision-factor hierarchy is a best practice that aids understanding by human decision-makers. Idiosyncratic representative style notwithstanding, it follows the ubiquitous form demonstrated by Saaty (1986). The elements in Figure 2 map to key DQ links from Figure 1.

- The rectangular box "PEMFC performance" at the top represents the *Goal or Focus*, in Saaty's (1986) terms. Thoughtful *framing*, in DQ terms, produce this aspect.

Figure 2: Hierarchy decomposition used for pipeline material-selection case study. (From Bu, et al (2026), Figure 5)



- The curve-cornered boxes *Criteria*. Corresponding to DQ's *values*, the range of measurable factors believed to materially distinguish between alternatives. For this case, they represent device physical-operating parameters *current density* "I", *pressure drop* "ΔP", and *Oxygen-uniformed index* "UI". These physical-operating parameters decisively influence the objective.

- The heptagonal shapes at the bottom depict decomposed crieteria. Saaty (1986) descrbes "...intermediate set of higher criteria that can both be decomposed into these attributes and are themselves decompositions of the higher level criteria or subcriteria...". In this case, each of three physical-device design specifications is believed to influence the operating parameters above.

Figure 2 omits the *discrete alternatives* from which the decision maker selects. Hierarchy depictions often include the alternatives when the set is small. Bu, et al (2026) however use a computational model of the device to simulate 36 distinct combinations of angle, width, and height. Using AHP to conduct analyze the risk of a large porfolio of distinct assets exemplifies another scenario in which the alterantives do not appear in the hierarchy.

The criterion hierarchy in Figure 2 is also idiosyncratic for another reason: Each of the physical-specification sub-criteria influences the operating-parameter criteria. Typically, constituent subcriteria are distinct to their associated criterion. Their representations appear as "clean" tree graphs. This excursion from common practice does not invalidate AHPs application, here. One might alternatively consider the *Analytic Network Process* (ANP) for scenarios like this (Saaty & Vargas (2013); Chen, et al (2019)).

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Comparative judgements about relative importance of criteria.

AHP imbues values and tradeoffs into decision-making through "...comparative judgments calls for setting up a matrix to carry out *pairwise comparisons* of the relative importance of the elements [criteria]..." (emphasis added) (Saaty (1986)). Practically, one assembles a list of unique, unordered criterion pairs for each hierarchic level. If the hierarchy for a given objective (or crietrion) identifies n criteria, then $\binom{n}{2}$ criterion pairs must be considered.

Transformation of comparative judments into *priority vectors*.

Entropy of edge-weighted, directed acyclic graphs.

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